

Who Leaves Teaching, and Why?

The Anatomy of Teacher Attrition in the United States, 1997–2024

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Abstract

One in six American teachers supposedly walks out the door every year. That number counts teachers who change schools and keep teaching; the share who actually leave the profession is half as large, and it is not rising. I measure it with one instrument pushed as far as it will go: the March Current Population Survey question comparing last year's longest job with this week's occupation, read from the raw Census files for twenty-eight consecutive years, 1997–2024. Leaving teaching averages 8.6 percent per year. It peaked at 10.3 percent in 2001 and at 10.2 percent in 2019, and stood at 7.7 percent in 2024, below its long-run mean. The anatomy of the flow explains its stability. Of every 100 leavers, 62 are not working the following March—38 of them past age 55, on their way out of the labor force for good—and only 26 hold another job. Booms do not lure teachers away: where exits respond to the business cycle at all, recessions push them out. Career changers face a gamble rather than a raise—the median switcher gains 15 log points, a quarter lose more than 30, and nearly half of all leavers earn nothing the following year. What predicts leaving is the contract, not the person: part-year work, part-time hours, and missing pension coverage dwarf every demographic. The one personal event that matters is a birth, which raises a woman's exit probability by six percentage points, twice the effect in nursing—and that margin has been fading for two decades as the motherhood exit disappears. Teacher pay slid from 80 to 72 percent of the median college graduate over the window; teaching is nonetheless the only comparison profession whose attrition fell. Teachers do leave—for retirement and for family, at the rate they always have. The exodus the debate keeps rediscovering is not in the data.

JEL codes: I20, J22, J31, J45, J63. **Keywords:** teacher attrition, occupational mobility, Current Population Survey, public-sector labor markets, teacher pay.

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1 Introduction

Every few years the United States rediscovers that its teachers are quitting. The discovery always arrives with a statistic—most recently, that one in six teachers walks away every year—and the statistic always carries the same message: this time it is an exodus. The concern behind the alarm is legitimate. Turnover is costly for schools and students (Ronfeldt et al., 2013; Hanushek et al., 2004), staffing is chronically hard in special education, mathematics, and high-poverty schools (Carver-Thomas and Darling-Hammond, 2017), and teacher pay has fallen visibly behind other college-graduate careers (Allegretto, 2023). But the statistic is not what it seems. Sixteen percent counts two very different behaviors as leaving: teachers who change schools and keep teaching, and teachers who leave the classroom altogether. Only the second group belongs in a sentence containing the word exodus—and that group has been seven to eight percent of the workforce for as long as any dataset can see.

This paper measures that group longer and deeper than it has been measured before. The March Current Population Survey asks every household two simple questions: what was the longest job you held last calendar year, and what do you do this week? Whoever taught last year and no longer teaches this March has left teaching within a window of at most fifteen months—and the survey then records where they landed (another occupation, unemployment, or out of the labor force) and what they earn. Harris and Adams (2007) introduced this design; Aldeman and Yi (2025) recently used its modern files to establish the level; the school-roster surveys of the National Center for Education Statistics confirm the same level independently (NCES, 2024). What no one has done is run the instrument at full depth and full length: every supplement from 1998 to 2025, read from the raw fixed-width Census files at the byte positions of the official technical documentation, harmonized into a single database of five million person-year records, validated person by person against within-panel classifications, decomposed by route, age, sex, sector, state, and family event, and priced by linking consecutive Marches. The result is a twenty-eight-year anatomy of who leaves teaching, where they go, and why.

The headline is that there is no trend. Leaving teaching averaged 8.6 percent over 1997–2024, with two nearly identical peaks a generation apart—10.3 percent in 2001, at the end of the dot-com boom, and 10.2 in 2019, on the eve of the pandemic—and a 2024 reading of 7.7, below the long-run mean (Figure 2). Covid produced a two-year excursion that fully reversed. The anatomy underneath explains why the rate is so stable, and the explanation is the story of this paper. Most leaving is not career change: of every 100 leavers, 62 are not working the following March—38 of them at ages 55 or older, the retirement route—and only 26 hold another job, a third of those still in education and care (Figure 3). The exits respond to life, not to the labor market. A new baby raises a woman’s probability of leaving by 5.9 percentage points, roughly double the effect in nursing or accounting (Figure 6)—though that margin has been closing for twenty-five years as the motherhood exit fades from American labor supply (Figure 7). A booming economy, by contrast, does not lure teachers away: across twenty-eight years and fifty-one jurisdictions, every statistically detectable cyclical response says recessions push teachers out, and none says booms pull them away (Figures 8 and 9). And leaving does not pay—or pays like a lottery. The median career changer

gains 15 log points, but a quarter lose more than 30, and 47 percent of all leavers report no earnings at all the following year (Figure 10). A risk-averse, pension-covered workforce buys few such tickets, which is precisely what the flat series shows. Meanwhile the pay of those who stay slid from 80 to 72 percent of the median college graduate (Figure 11)—and teaching is nonetheless the only comparison profession whose attrition fell between the first and second halves of the sample (Figure 13).

The paper is organized around its figures. Section 2 describes the instrument, the pooled database, the definitions that reconcile the numbers in the public debate, and the validation exercises. Sections 3–10 present the anatomy: the changing workforce, the rate, the routes, the family margin, the cycle, pay, the multivariate portrait, and the comparison professions. Section 11 concludes with what the anatomy implies for policy. Appendices carry the instrument audit, the comparison with Harris and Adams (2007), construction details, robustness tables, and additional figures.

2 Data and measurement

2.1 The March retrospective instrument

The Annual Social and Economic Supplement (ASEC) of the CPS, fielded every March, asks two questions that together identify profession leaving. The first is retrospective: the occupation of the longest job held during the previous calendar year (OCCUP). The second is contemporaneous: the occupation of the job held this week (PEI00CC), together with the labor force status recode (A_LFSR), which distinguishes the employed, the unemployed, and those outside the labor force. A teacher is a respondent whose longest job last year carried a K–12 teaching occupation code (preschool through secondary and special education, plus the “other teachers” residual; postsecondary teachers, aides, and tutors are excluded). A leaver is a teacher who, by the March interview, is not employed in a teaching occupation: working in some other occupation (a switcher), unemployed, or out of the labor force. Because status comes from A_LFSR rather than from the occupation variable—which carries the last job held even for the unemployed—the three routes are mutually exclusive and exhaustive.

Three features of the design matter for interpretation and recur throughout the paper. **Timing.** The supplement of survey year $t+1$ measures calendar year t ; the exit itself occurs at some point between the end of the teaching spell in year t and the March interview—for teachers, typically at the end of a school year. I therefore date exits by the calendar year taught and, where the labor market at the moment of observation matters (Section 7), measure it in the survey month itself. **Leave is not exit.** A teacher on maternity or medical leave in March reports “with a job, not at work,” counts as employed, and—if the job is a teaching job—as a stayer. The leavers counted here are genuine separations. **Movers are invisible by design.** A teacher who changes school but keeps teaching keeps the occupation code and is a stayer. This is precisely what makes the CPS number a measure of profession attrition rather than school turnover.

2.2 One pooled database from the raw files

All results come from a single person-level database built from the original Census public-use files: 5,009,129 person-year records across the 28 supplements of survey years 1998–2025, containing 104,545 teacher-year observations (85,497 for college graduates aged 18 and over, the baseline universe). For survey years 2011–2025 the modern machine-readable files are used directly. For survey years 1998–2010 the public files are fixed-width records whose layouts are not machine-readable; every variable was extracted at the byte positions published in the official Census technical documentation (`cpsmar98.pdf`–`cpsmar10.pdf`), so that education, supplement weights, labor force status, demographics, household structure, last-year job characteristics (weeks, hours, class of worker, pension coverage), earnings, and state of residence exist on a common footing in every year. Occupation codes are harmonized across the four Census classification regimes with era-consistent code sets. Appendix C documents the construction; Table 3 reports record counts by source. All series are weighted with the ASEC supplement weight unless noted.

2.3 What “leaving” means: reconciling the public numbers

The headline numbers in the policy debate are consistent with one another once definitions are aligned; the alarm comes from mixing them. The NCES Teacher Follow-up Survey reports total turnover of public school teachers of 13–16 percent across its waves. That total is the sum of two components the survey itself keeps separate: movers, who change school but keep teaching (5–8 percent per wave), and leavers, who exit the profession (5.6 percent in 1988–89, rising to 7.4–8.4 percent from 2000–01 onward) (NCES, 2024; Tan et al., 2025). The CPS estimate of roughly 7.6 percent in Aldeman and Yi (2025) measures leavers only, as does the 7.7 percent of Harris and Adams (2007) for 1992–2001, and as does the series built here, which averages 8.2 percent for college-graduate teachers over 2015–2024 and 8.6 percent over the full window. There is no contradiction anywhere in this literature: school-to-school mobility roughly doubles profession leaving, and only the latter belongs in a sentence containing the word exodus. The estimates here sit slightly above the NCES leaver rates for overlapping years, as they should: the CPS covers private school teachers, whose attrition is nearly twice the public rate (Section 5), while the NCES figure is public-only.

2.4 Validation

Two audits back the instrument. First, a person-level cross-validation: for respondents observed in the CPS monthly rotation long enough to be classified by a four-month within-panel criterion and who also answer the March retrospective question, the two classifications agree in 99.1 percent of cases ($n = 9,461$; Appendix A, Table 1). Second, a timing audit: the correlation of the leaving series with the national unemployment rate is +0.08 using the year before the teaching year, +0.32 using the teaching year itself, and +0.47 using the year the exit is observed—exactly the gradient the retrospective design implies, and



Figure 1: The changing profile of the teaching workforce, 2005–2024.

College-graduate teachers aged 18+ whose longest job last year was K–12 teaching; ASEC supplement weights. Thin gray lines are annual values; blue lines are centered 3-year averages, with endpoint values printed. “Public school,” “part-time,” and “pension plan” refer to the longest job held last year; the open circle marks 2013, a year with a reduced sample due to the ASEC redesign. Source: author’s calculations from March CPS supplements, survey years 2006–2025.

evidence that the instrument dates transitions where it claims to. External anchors point the same way: the NCES leaver rates for its five overlapping waves fall within a point of the series.

3 A changing profession

Before following anyone out the door, it is worth looking at who is in the room. Figure 1 tracks the composition of the teaching workforce over the last two decades, panel by panel; each panel shows the annual series in light gray and a three-year average in blue. Four slow-moving facts stand out, and each returns later in the paper as a mechanical driver of attrition.

The workforce is aging: mean age rose from 42.8 to 44.8, and the share of teachers aged 55 or more rose with it—which matters because Section 5 shows the 55+ retirement route is the largest single component of attrition. It is increasingly credentialed (45 to 51 percent with a master’s degree or more) and slowly diversifying (8 to 11 percent Black). The share with a young child at home has fallen from 19 to 14 percent, the demographic echo of later and lower fertility—and, as Section 6 shows, the family exit route has fallen even faster than the share of potential mothers. Finally, the employment contract has deteriorated on the one margin the attrition model will single out: pension coverage at the last-year job fell from 80 to 66 percent, the steepest change in any panel.

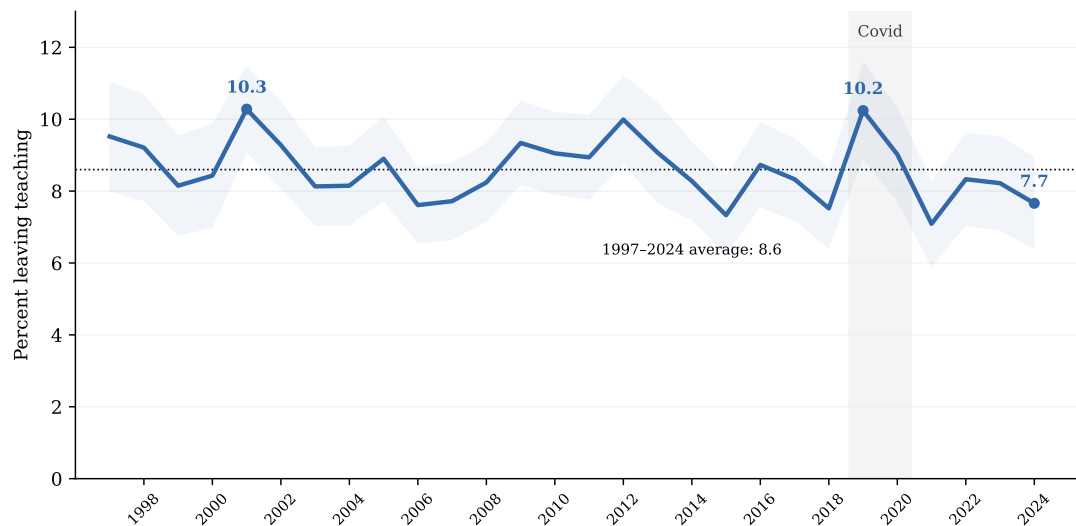


Figure 2: Leaving teaching, 1997–2024.

Share of college-graduate teachers (longest job last calendar year) not employed in a teaching occupation the following March, by route-based definition; ASEC weights. Shaded band: 95 percent confidence interval using a design factor of 1.5. Dotted line: the 1997–2024 mean (8.6). The gray band marks the Covid supplements. The occupation classification changes between calendar 2001 and 2002 (1990 to 2000 Census codes); a core-codes-only series shows the same profile (Appendix D). Source: author’s calculations from March CPS supplements, survey years 1998–2025.

4 The rate: twenty-eight years without a trend

Figure 2 is the paper’s spine: the share of college-graduate teachers who left the profession each calendar year from 1997 to 2024, on one continuous, consistently defined, weighted series—to my knowledge the longest ever assembled from this instrument.

Read it the way a policymaker should. The level: 8.6 percent on average, never below 7, above 10 only three times in twenty-eight years. The trend: there is none. The two highest points sit eighteen years apart, one at the end of a boom and one on the eve of a pandemic, and the series ends below its mean. Covid, the event that launched a thousand exodus headlines, reads as an excursion, not a break: 10.2 in 2019, 9.0 in 2020, 7.1 in 2021, back to trend since. An annual outflow of eight percent, composed overwhelmingly—as the next section shows—of retirements, family spells, and early-career sorting, is best understood as the profession’s natural rate of attrition: the turnover a five-million-person, majority-female, pension-covered occupation generates in the ordinary course of life-cycle events. Fluctuations around it have never exceeded two points in either direction. Any policy narrative that begins from an accelerating exodus begins from a fact the data do not contain.

5 Where leavers go

So eight of every hundred teachers leave each year. Where do they go? Figure 3 follows them. Of every 100 college-graduate teachers who leave, 62 are not working the following March: 38 are out of the labor force at ages 55 or older—the retirement route—24 are out of the labor force younger than 55, and 12

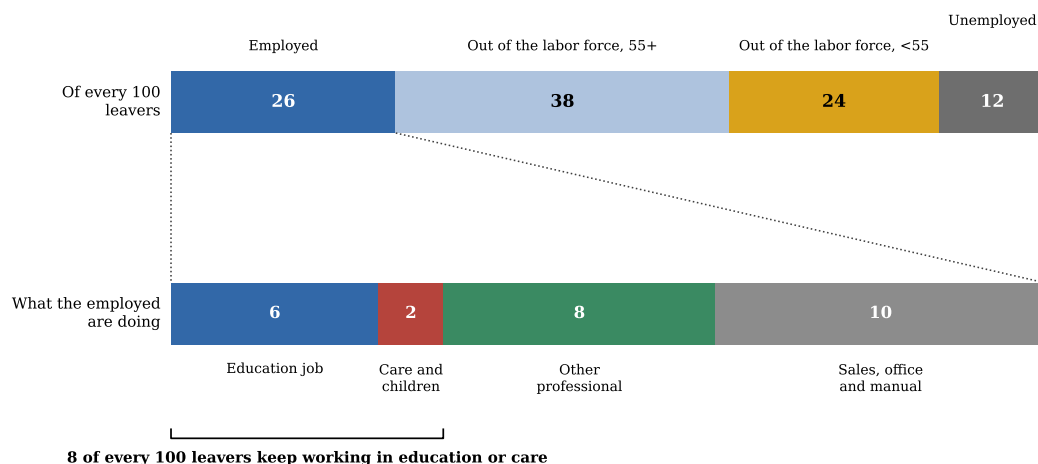


Figure 3: Of every 100 teachers who leave the profession.

College-graduate leavers pooled over calendar 2015–2024, ASEC weights. Top bar: labor force status the March after the teaching year; “out of the labor force” is split at age 55. Bottom bar: the occupational destination of the 26 employed leavers, rescaled to sum to their total. Source: author’s calculations from March CPS supplements.

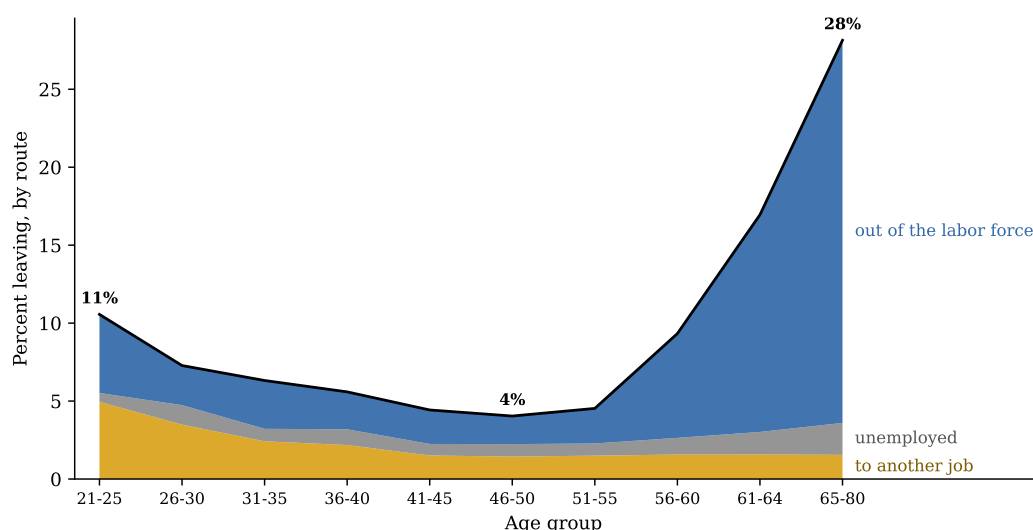


Figure 4: Leaving by age, decomposed by route.

College-graduate teachers pooled over calendar 2015–2024, ASEC weights; ten age groups. The black line is the total leaving rate; stacked areas decompose it into exits to another job (gold), unemployment (gray), and out of the labor force (blue). Source: author’s calculations from March CPS supplements.

are unemployed. Only 26 hold a job. And of those 26, eight remain in education or care work—school administration, counseling, childcare—so that barely one leaver in six matches the scenario the exodus narrative describes: a working-age teacher productively employed outside education a year later.

The age profile in Figure 4 shows where each route lives. Total leaving traces the classic U: 11 percent among teachers aged 21–25, a flat 4–5 percent through ages 41–55, then 9, 17, and 28 percent in the three oldest groups. The decomposition explains both arms. The left arm is career search: nearly half of the exit of the youngest teachers is to another job, a share that decays rapidly with age—and, notably, the early hump is a male phenomenon (15 percent for men aged 21–25 versus 9 for women). The right arm is

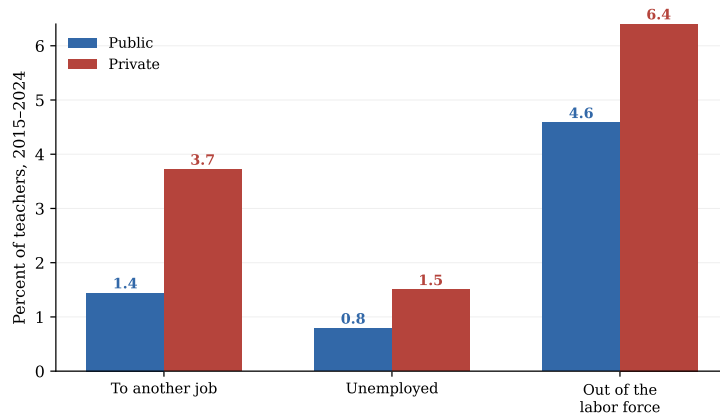


Figure 5: Public and private school teachers leave differently.

Route-specific annual exit rates of college-graduate teachers by sector of the last-year job (class-of-worker: government vs. private), pooled 2015–2024, ASEC weights. A private-school teacher who moves to a public school—or vice versa—remains a teacher and is not a leaver here. Source: author’s calculations from March CPS supplements.

retirement, almost entirely: from age 55 the out-of-labor-force route accounts for nine tenths of a rapidly rising total, while the job-switching route is flat at 1.5 percent from age 35 to the end. Unemployment never exceeds two percent at any age. Mid-career teachers—the group schools most need to retain, and the group the crisis narrative implicitly describes—are the least mobile workers in the figure.

One split deserves its own figure. Private school teachers leave the profession at nearly twice the public rate—11.6 versus 6.8 percent a year—and the excess runs through every route (Figure 5): to other jobs (3.7 vs. 1.4), to unemployment (1.5 vs. 0.8), and out of the labor force (6.4 vs. 4.6). Because sector switching within teaching does not count as leaving, this is genuine profession attrition, not churn between sectors. The gap is persistent across the whole window and concentrated among the young: a private school loses roughly one in five of its under-30 teachers to the profession every year. The public sector premium in retention—tenure protections, salary schedules, and above all pension coverage—previews the multivariate result of Section 9.

6 The family margin, and its slow disappearance

If the contract is the main systematic predictor of exit, the main personal event is a birth. In the pooled model of Section 9, a new baby raises a teacher’s probability of leaving by 4.7 percentage points; among women the effect is 5.9 points. Figure 6 puts that number in comparative perspective: the teaching effect is roughly double the birth-year exit in nursing (2.6) and accounting (3.5), the two comparison professions where the effect is also statistically detectable.

Why teaching? Two readings coexist, and the data support both. The harsher one is rigidity: teaching offers no part-time ladder, no shift swaps, and no remote option—a teacher either covers her classroom at its hours or does not work—so the arrival of an infant forces a binary choice that nursing, with its shift structure, does not force. The gentler one runs the logic of career interruption costs in Goldin (2014) in

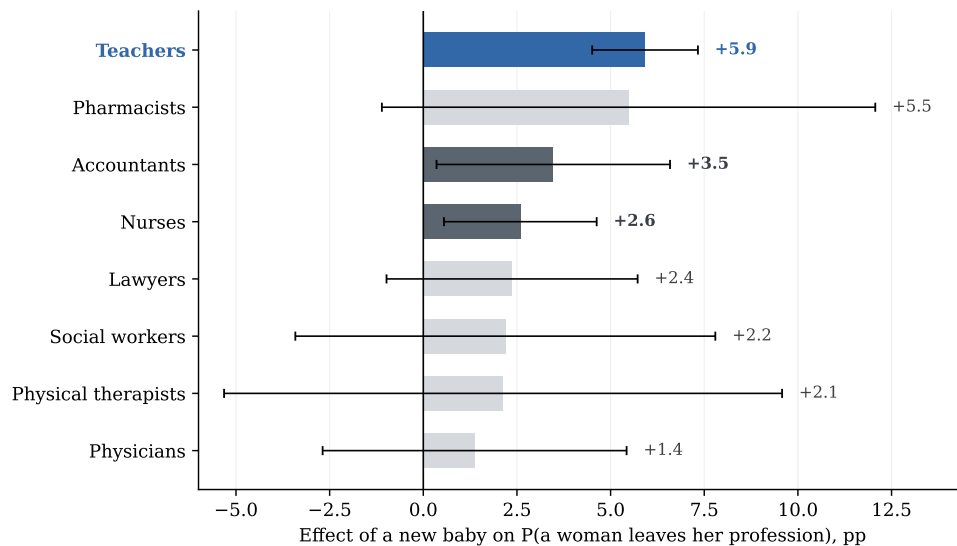


Figure 6: The birth-year exit, teaching versus seven professions.

Average marginal effect of a new baby (own child aged 0 in the household) on the probability that a woman leaves each profession, from profession-specific probits with age, age squared, marital status, education, and year effects; whiskers are 95 percent confidence intervals. Solid bars are significant at 5 percent; light bars are not. Samples pool 2010–2024. Source: author’s calculations from March CPS supplements.

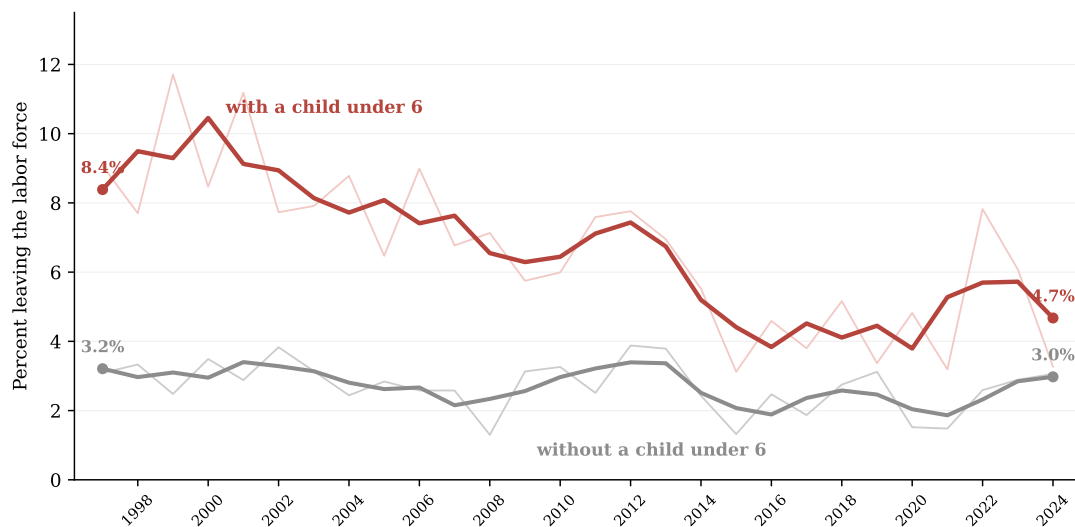


Figure 7: The fading motherhood exit, 1997–2024.

Women teachers aged 22–45, college graduates; annual probability of leaving the labor force (thin lines) and centered 3-year averages (bold), separately for those with and without an own child under 6 at home. Endpoint values printed. Source: author’s calculations from March CPS supplements, survey years 1998–2025.

reverse: teaching is the college profession where stepping out is cheapest—certification persists, seniority premia are flat, jobs exist everywhere—so mothers who in another career would hang on through the infant years can afford to pause. A linked-sample audit supports the second reading as a substantial component: about a third of new-mother leavers are observed teaching again within a year. The birth effect, in other words, is partly a revolving door, not only an exit.

Whatever its interpretation in the cross-section, the margin is disappearing in the time series. Figure 7

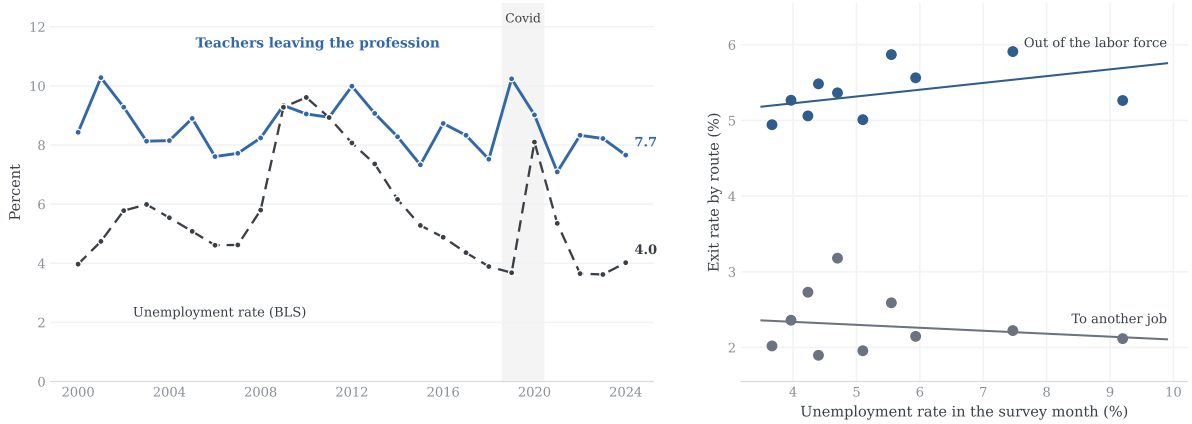


Figure 8: Leaving teaching and the labor market, 1997–2024.

Left: the leaving series of Figure 2 (annual, from calendar 2000) against the BLS national unemployment rate (annual average). Right: binned scatter (deciles of the unemployment rate in the survey month, March of $t+1$, when the exit is observed) of the two labor-market routes, with OLS fits on the underlying annual data; correlations are -0.16 (to another job) and $+0.29$ (out of the labor force). Unemployment: BLS via FRED (UNRATE). Source: author’s calculations from March CPS supplements.

tracks the probability that a young woman teacher leaves the labor force, separately for mothers of children under six and for other women. In the late 1990s the gap was five to seven points a year—the classic exit to home. It has narrowed almost continuously for twenty-five years and stands below two points at the end of the window (4.7 versus 3.0 in 2024). Combined with the falling share of teachers who have young children at all (Figure 1), the family route now explains a shrinking slice of total attrition—the composition shift visible in Appendix Figure 14. A state-level test of whether paid family leave programs accelerate this decline is underpowered but points the right way: the birth-year exit effect is 2.1 points smaller in state-years with an active program, on only 268 treated births (Appendix E).

7 Attrition and the business cycle

A popular version of the exodus story is cyclical: when the economy booms, outside options improve and teachers are poached. The data say close to the opposite. In levels (Figure 8, left), leaving is a flat line while unemployment swings through three recessions; the two series touch only around Covid. In the cross-section of years (right panel), the job-switching route is essentially acyclical—if anything it falls when unemployment is high ($r = -0.16$), as fewer outside offers arrive—while the withdrawal route rises moderately in downturns ($r = +0.29$): slack markets bring forward retirements and returns to home production, an old pattern in female labor supply. The route into unemployment correlates $+0.55$ with the aggregate rate, a near-mechanical validation of the instrument.

State-level variation sharpens the asymmetry. Estimating the within-state relationship between leaving and the state unemployment rate for all fifty-one jurisdictions over twenty-eight years yields a distribution of slopes centered near zero—in the large states, where power is real, the estimates are tight zeros (California $+0.21 \pm 0.22$)—with a telling tail: the only four states where the relationship is statistically distinguishable from zero (Mississippi, Arizona, Maine, Indiana) all lie on the “exits rise in downturns” side, and not one

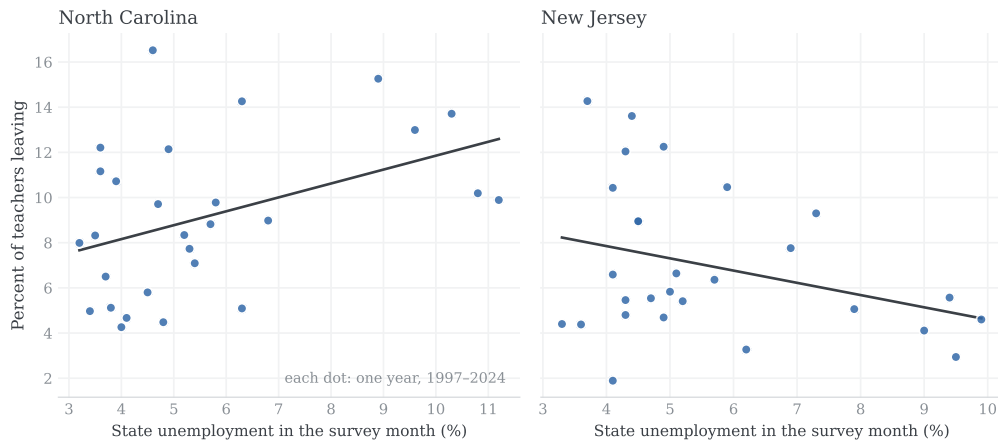


Figure 9: Two illustrative states.

Annual state leaving rates of college-graduate teachers (dots; one per year, 1997–2024) against the state unemployment rate in the survey month (BLS Local Area Unemployment Statistics), with OLS fits. North Carolina and New Jersey are shown as examples of the two signs; the full 51-state distribution of slopes is in Appendix Figure 15. State of residence is recovered from the household records of the raw files (Appendix C). Source: author’s calculations from March CPS supplements.

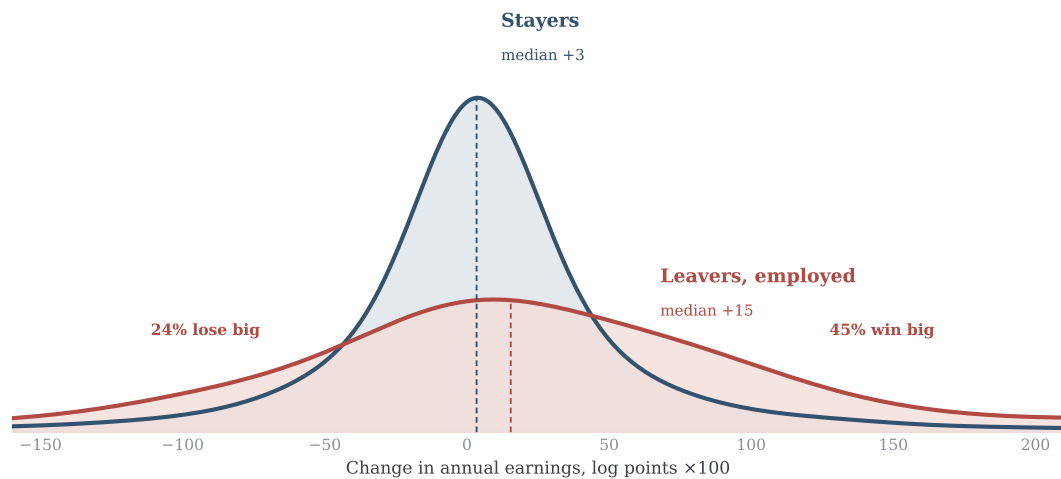


Figure 10: The earnings change of leaving, person by person.

Kernel densities of the person-level change in annual wage and salary earnings (log points $\times 100$) between the teaching year and the next, for teachers linked across consecutive supplements (rotation-group links, survey years 2011–2025; identity validated on sex and age). Navy: stayers (12,195 links; median +3). Red: leavers employed the following year (225 links; median +15). “Lose big”/“win big” denote changes beyond ± 30 log points. The sample conditions on positive earnings in both years; 47 percent of all leavers report zero wage and salary earnings the following year. Bandwidth 0.25. Source: author’s calculations from March CPS supplements.

state shows a significant “exits rise in booms” pattern (Appendix Figure 15). Figure 9 illustrates the two signs with one state each. Where teacher exits respond to the cycle at all, recessions push; booms do not pull.

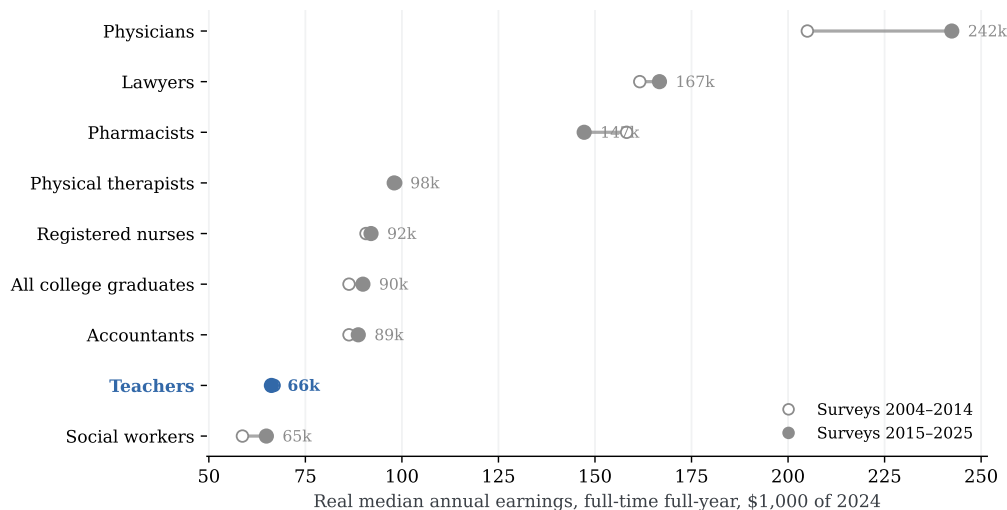


Figure 11: Real pay by profession, two symmetric windows.

Weighted median annual wage and salary earnings of full-time, full-year college graduates aged 25–60, in each profession and overall, deflated to 2024 dollars with the CPI-U; the two windows pool survey years 2004–2014 and 2015–2025. Filled dots (with values): recent window. Occupation codes harmonized across classification regimes. Source: author’s calculations from March CPS supplements; CPI: BLS via FRED.

8 Pay: the gamble of leaving and the slide of staying

Does leaving pay? The honest answer is that it can, and that is different from it does. Linking teachers across consecutive Marches prices each transition at the person level, and Figure 10 draws the result. Stayers experience ordinary wage growth: a median of +3 log points, tightly distributed. Leavers who land in another job show a median of +15—a real premium—but around that median the distribution is a flattened tent with mass far into both tails: 24 percent lose more than 30 log points, 45 percent gain more than 30. And the figure conditions on earning anything at all. Forty-seven percent of leavers report zero wage and salary earnings the following year, consistent with the route decomposition of Figure 3: most leavers were not walking into better-paid jobs, because most leavers were not walking into jobs. Leaving teaching for money is a lottery with a modest median prize. The rational response of a risk-averse, pension-covered workforce is to buy few tickets—and that is what Figure 2 shows.

The staying side of the ledger has deteriorated. In real terms, the median full-time, full-year teacher earns \$66,000 (2024 dollars) in the recent decade—essentially unchanged from the decade before—while registered nurses moved to \$92,000, accountants to \$89,000, and the median college graduate to \$90,000 (Figure 11). Indexed to 2002, teacher real pay ends the window six percent below its starting point, against +3 for nurses and +8 for accountants (Appendix Figure 17); as a ratio, teacher pay slid from 80 to 72 percent of the median college graduate. This is the CPS version of the teacher pay penalty documented by Allegretto (2023), measured here with one consistent instrument alongside everything else in the paper. The juxtaposition with Section 10 is the point: the profession whose relative pay fell furthest is also the only one whose attrition declined. Pay operates on margins the exit rate hides—who enters, who feels trapped, how retention interacts with pensions—not as a simple valve on the outflow.

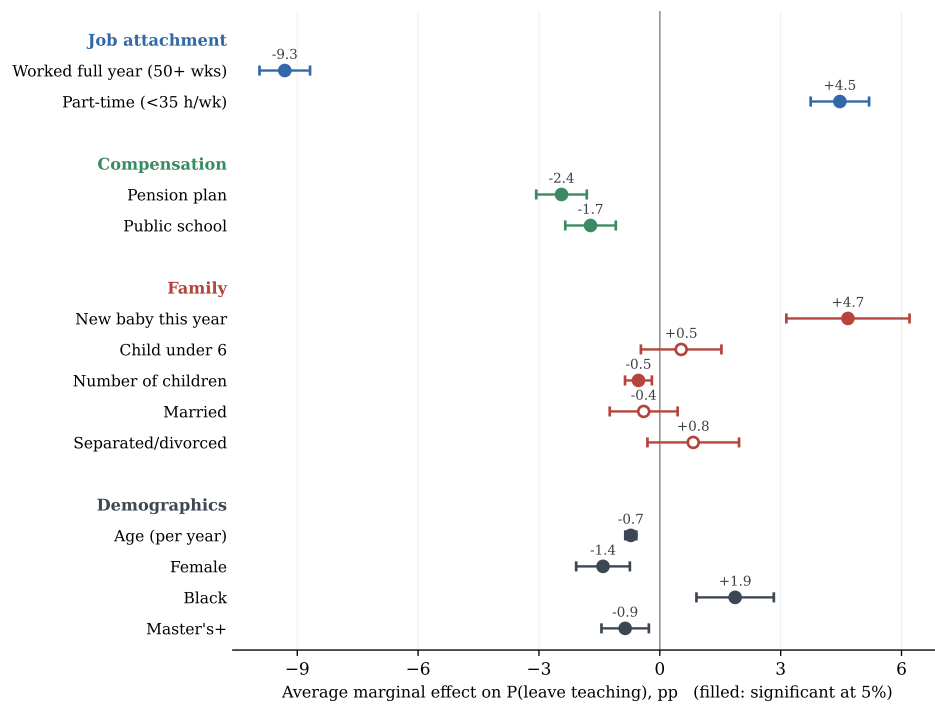


Figure 12: Who leaves teaching: probit average marginal effects.

Average marginal effects (percentage points) on the probability of leaving teaching, from a probit on college-graduate teachers pooled over calendar 2015–2024 with year effects; whiskers are 95 percent confidence intervals; filled markers are significant at 5 percent. “Worked full year” is 50+ weeks at the last-year job; “part-time” is under 35 weekly hours. $n = 28,900$. Robustness across estimators, weights, fixed effects, windows, and definitions, and subsample estimates, are in Appendix E. Source: author’s calculations from March CPS supplements.

9 Who leaves: the multivariate portrait

Figure 12 assembles the portrait in one model, and the model tells one story: the contract, not the person. A teacher who worked the full year is 9.3 points less likely to leave; part-time hours add 4.5 points; pension coverage subtracts 2.4; a public school subtracts 1.7. Against these, demographics are small: each year of age subtracts 0.7 (the U-shape enters through age squared), women are 1.4 points less likely to leave than men, Black teachers 1.9 more, a master’s degree subtracts 0.9. The family block is a tale of one variable: the number of children and the presence of a child under six do essentially nothing, but the birth year itself adds 4.7 points—the event, not the state, drives the exit.

The appendix battery shows the portrait is not an artifact of specification. Across probit, weighted LPM, logit, state-and-year fixed effects, the full 1997–2024 window ($n = 85,430$), and the occupation-pair definition of leaving, the job-attachment and birth coefficients never lose significance or change sign (Table 5). The subsample estimates (Table 6) add two findings worth text: a child under six raises women’s exit (+1.3) but lowers men’s (–2.2)—household specialization in one row—and pension coverage retains men more than twice as strongly as women (–4.5 versus –2.0).

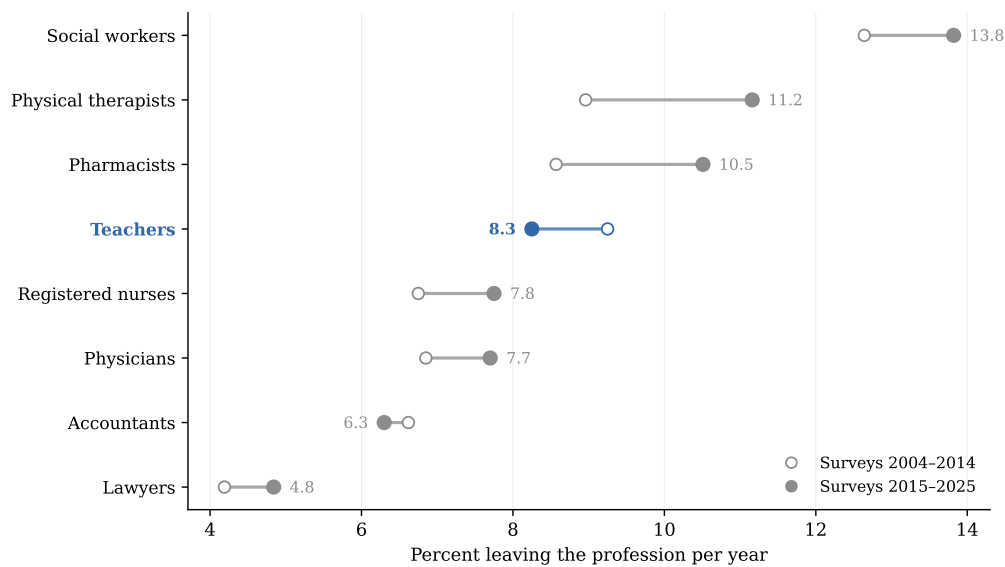


Figure 13: Occupational leaving in eight professions, two symmetric windows.

Annual rate of leaving each profession (same route-based definition as for teachers, college graduates, ASEC weights), pooled over survey years 2004–2014 (open) and 2015–2025 (filled, with values). Occupation codes harmonized across the 2002, 2010, and 2018 Census classification regimes. Source: author’s calculations from March CPS supplements.

10 Teaching among the professions

Is an eight percent outflow a lot? Figure 13 answers with the only benchmark that means anything: other professions, measured identically. Teaching sits in the middle of the college-professional range—above accountants and lawyers (5–6), essentially tied with nurses and physicians (8), far below social workers (14) and the therapy professions (11). Two things elevate the comparison beyond a ranking. First, teaching is the only profession in the figure whose leaving rate fell between the two windows (9.3 to 8.3); every other profession rose. Second, the comparison disciplines interpretation: whatever is pushing pharmacists and physical therapists up is economy-wide, not a crisis of schools. The annual series behind these windows are in Appendix Figure 16, and the same comparison estimated on Harris and Adams’s 1992–2001 window is in Appendix B—the ordering of professions is the same three decades later.

11 What the anatomy means for policy

This paper set out to measure, not to prescribe; but a measurement this long constrains prescriptions, and it is worth stating how.

Retire the exodus frame. Profession leaving is flat at its generation-long level, was flat through Covid within two years, and is lower today than in 2001. Policies premised on an accelerating outflow—emergency certification pipelines, across-the-board retention bonuses—address a trend that does not exist, and their evaluation against a mythical baseline will always look like success. The turnover problem American schools do have is distributional—by school, subject, and district (Carver-Thomas and Darling-Hammond,

2017; Hanushek et al., 2004)—and the CPS, which sees professions rather than schools, is the wrong lens for it; but a national policy sold on the national rate is answerable to the national rate.

Aim at the margins that move. The anatomy locates three. The retirement route (38 of every 100 leavers) responds to pension rules, and its mean age has already drifted from 61.3 to 65.2 across the window—policies that manage the timing of exits (phased retirement, return-to-work rules) operate on the largest single flow. The family route responds to the compatibility of teaching with an infant at home: the birth-year exit is twice nursing’s, a third of it is a revolving door already, and the fading of the motherhood exit shows the margin is elastic to social change; job-protected paid leave and genuine part-time ladders are the obvious levers, and the underpowered but right-signed paid-leave interaction estimated here suggests the experiment is worth running properly. The early-career route (11 percent at ages 21–25, half of it to other jobs, disproportionately male and private-sector) is standard career search; the productive response is selection and early support, not universal retention spending on a flow that mostly self-corrects by age 30.

Read the pay problem correctly. Nothing here says pay does not matter; the point is where it matters. The exit rate cannot fall much—it is already anchored by routes pay does not touch—so the return to raising relative pay lies elsewhere: in who chooses to enter, in the quality margin of who stays, and in fairness to a workforce whose real median earnings have not grown in twenty years while every comparison profession’s did. The pension result carries a specific warning: coverage retains (2.4 points, twice that for men), and coverage has fallen 14 points in two decades. Letting the deferred part of the compensation package erode while debating the current part is, on this evidence, exactly backwards.

Treat teaching’s turnover as the price of its accessibility. The comparison figures reframe the profession’s churn: teaching pays 72 cents on the college dollar, yet loses fewer of its members than social work, therapy, or (now) pharmacy, and uniquely improved across decades. Part of the reason is the same flexibility that shows up as the birth-year spike and the revolving door: teaching is cheap to leave and cheap to re-enter, which makes it the college profession most compatible with an interrupted life. A policy agenda that treated that accessibility as an asset to be organized—formal re-entry paths, leave that preserves attachment, part-time roles that are careers rather than exits—would work with the grain of the anatomy documented here.

The contribution of this paper is the anatomy itself: one instrument, twenty-eight consecutive years, every exit routed, priced, and modeled. On that evidence, the American teaching profession is not hemorrhaging. It is aging, underpaid relative to its peers, uniquely exposed to the arithmetic of family life, and steady—losing its members to the same life events it has always lost them to, at the same rate, while the conversation about it swings with each news cycle. Policy should be built on the steadiness, not the swings.

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A The instrument, audited

For respondents in CPS rotation groups that yield four consecutive monthly interviews after the reference year together with a March retrospective response, profession leaving can be classified two independent ways: by the within-panel criterion (teaching occupation in all four post months versus none) and by the March retrospective question. Table 1 cross-tabulates the two for the pooled linked sample. Among panel-classified stayers, 96.8 percent are retrospective stayers; among panel-classified leavers, the March question recovers the exit or the non-teacher status in 99.2 percent of cases; overall agreement is 99.1 percent. The March instrument, in short, reproduces at the person level the classification a short panel would give—while covering four times the sample and every year since 1998.

Table 1: March retrospective vs. linked-panel classification.

Linked-panel classification	March retrospective classification			
	Not a teacher	Stayed	Left	Total
Stayer (teaches all 4 post months)	198	7,911	67	8,176
Leaver (teaches in none)	1,018	10	257	1,285
Total	1,216	7,921	324	9,461

Pooled linked sample, survey years 2011–2024. Rows: 4-month within-panel classification; columns: March retrospective classification. Source: author’s calculations from CPS basic monthly files and March supplements.

B Comparison with Harris and Adams (2007)

Harris and Adams pooled the March supplements of survey years 1992–2001 and reported route-decomposed leaving for teachers, nurses, social workers, and accountants. Table 2 reproduces their estimates alongside ours for 2015–2024, constructed identically. Levels are within 1.5 points for every profession, the ordering is unchanged after a quarter century, and the route structure—teachers leaving mostly to non-employment, social workers mostly to other jobs—is intact. The stability of this comparison across thirty years and two occupational classification systems is itself a finding: the relative anatomy of professional turnover is a slow-moving object.

Table 2: Leaving rates and routes: 1992–2001 vs. 2015–2024.

	Harris & Adams, 1992–2001				This paper, 2015–2024			
	All	Switch	Unemp.	Out LF	All	Switch	Unemp.	Out LF
Teachers	7.73	2.59	0.61	4.53	8.25	2.12	1.00	5.12
Nurses	6.09	1.68	0.86	3.54	7.66	3.31	0.80	3.54
Social workers	14.94	10.87	1.34	2.73	13.55	9.45	1.22	2.89
Accountants	8.01	4.10	1.55	2.36	6.30	1.79	1.18	3.32

Left block: Harris and Adams (2007), Table 1, March CPS 1992–2001. Right block: author’s calculations, identical definitions, March CPS 2015–2024 (survey years 2016–2025). All rates in percent per year.

C Data construction

Sources. Survey years 2011–2025: Census machine-readable ASEC person files. Survey years 1998–2010: the original fixed-width public-use files, parsed at the byte positions published in the official technical documentation (`cpsmarYY.pdf`); positions are constant within the 1998–2002 and 2003–2010 layout eras. Household-level variables (state of residence) are read from the household records and merged on the household sequence number. Every extraction is validated with sanity gates (age distributions, weight positivity, education recode ranges, teacher shares of occupation) that abort rather than ship on failure.

Table 3: Record counts by source.

Source	Survey years	Persons	Teacher-years	Teacher-years (BA+, 18+)
Machine-readable ASEC files	2011–2025	2,585,797	54,620	45,296
Fixed-width files, official layouts	1998–2010	2,423,332	49,925	40,201
Total	1998–2025	5,009,129	104,545	85,497

Teacher-years: longest job last year in a K–12 teaching occupation. The baseline analysis universe is college-graduate teachers aged 18+. Calendar-year coverage is 1997–2024.

Occupation codes. K–12 teaching is defined as codes 155–159 (1990 classification; surveys through 2002), 2300–2340 (2002 and 2010 classifications), and 2300–2330 plus 2360 (2018 classification; surveys from 2020), excluding postsecondary teachers, aides, and tutors. A core-codes-only series (excluding the “other teachers” residual in all years) shows the same level and profile. Comparison professions use analogously era-consistent sets.

External series. National unemployment: BLS via FRED (UNRATE); state unemployment: BLS Local Area Unemployment Statistics via FRED, March values of the survey year; deflator: CPI-U annual averages via FRED, base 2024. All cached with their sources in the replication package.

D Additional figures

E Appendix tables

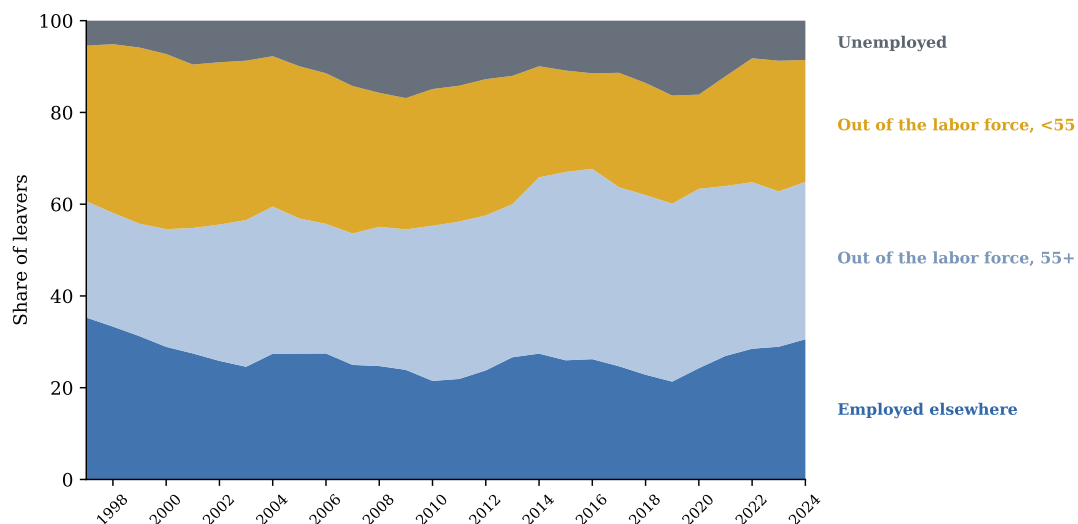


Figure 14: Route composition of leavers, 1997–2024.

Shares of college-graduate leavers by route (3-year averages, renormalized to 100). The retirement-age share rises from roughly a quarter to nearly forty percent; the young out-of-labor-force (family) share falls symmetrically. Source: author’s calculations from March CPS supplements.

Table 4: Sample balance: teachers and other college graduates, 2015–2024.

	Teachers, women	Teachers, men	All teachers	Other college workers
<i>Demographics</i>				
Age (years)	44.2 (12.8)	44.9 (13.3)	44.3 (12.9)	44.1 (13.6)
Female	100.0	0.0	76.0	48.7
Black	9.6	9.5	9.6	9.6
Married	66.6	67.7	66.9	62.6
Child under 6	16.1	17.2	16.4	15.0
Number of children	0.8 (1.1)	0.8 (1.1)	0.8 (1.1)	0.6 (1.0)
<i>Education</i>				
Master’s degree or higher	51.3	53.6	51.9	35.4
<i>Job held last year</i>				
Government employer	70.3	70.4	70.3	15.5
Part-time (<35 h/wk)	15.2	11.4	14.3	13.5
Full year (50+ weeks)	71.1	74.8	72.0	85.0
Pension plan at work	66.4	67.3	66.6	46.1
Wage earnings, median (\$)	50,000	58,000	52,000	70,000
<i>Transitions</i>				
Leaves teaching within a year	7.9	9.1	8.2	–
Person-years (unweighted N)	22,097	6,785	28,882	277,364

Weighted means; standard deviations in parentheses for continuous variables. Wage earnings are medians, current dollars, conditional on positive earnings. “Leaves teaching” is undefined for non-teachers. Source: author’s calculations from March CPS supplements.

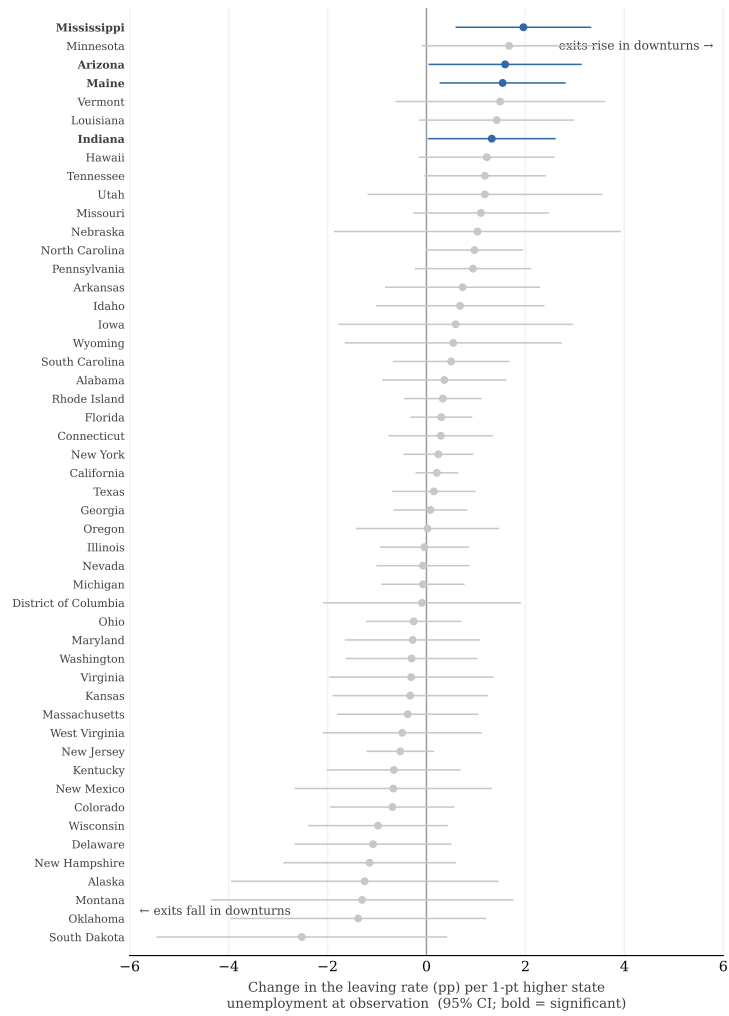


Figure 15: State-level cyclicalities of leaving: all 51 jurisdictions.

Slope of a weighted linear probability model of leaving on the state unemployment rate in the survey month, estimated separately by state on all years 1997–2024; 95 percent intervals, heteroskedasticity-robust. Bold: significant at 5 percent. Source: author's calculations from March CPS supplements and BLS LAUS.

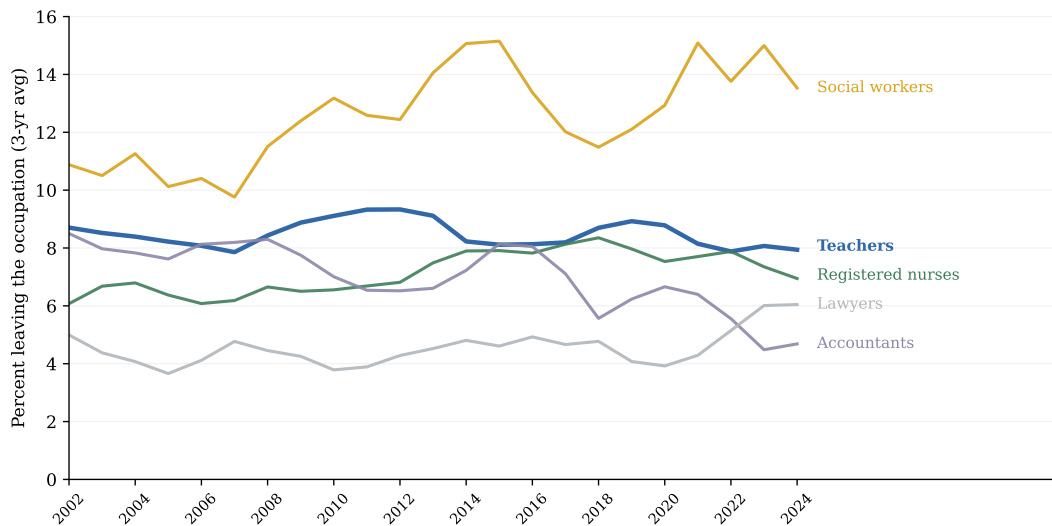


Figure 16: Occupational leaving in five professions, annual series.

Three-year averages of the annual occupational leaving rate, college graduates, identical definitions across professions; the window figures of the main text pool these series. Source: author's calculations from March CPS supplements, survey years 2003–2025.

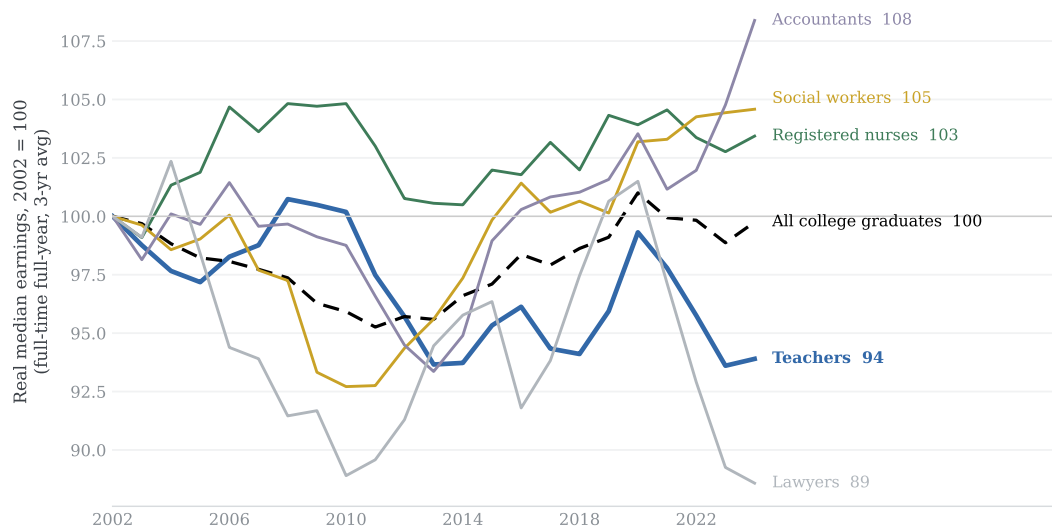


Figure 17: Real median pay by profession, indexed.

Real median annual earnings of full-time full-year college graduates aged 25–60 by profession, CPI-U deflated, indexed to 2002 = 100, 3-year averages; endpoint values printed. Source: author's calculations from March CPS supplements.

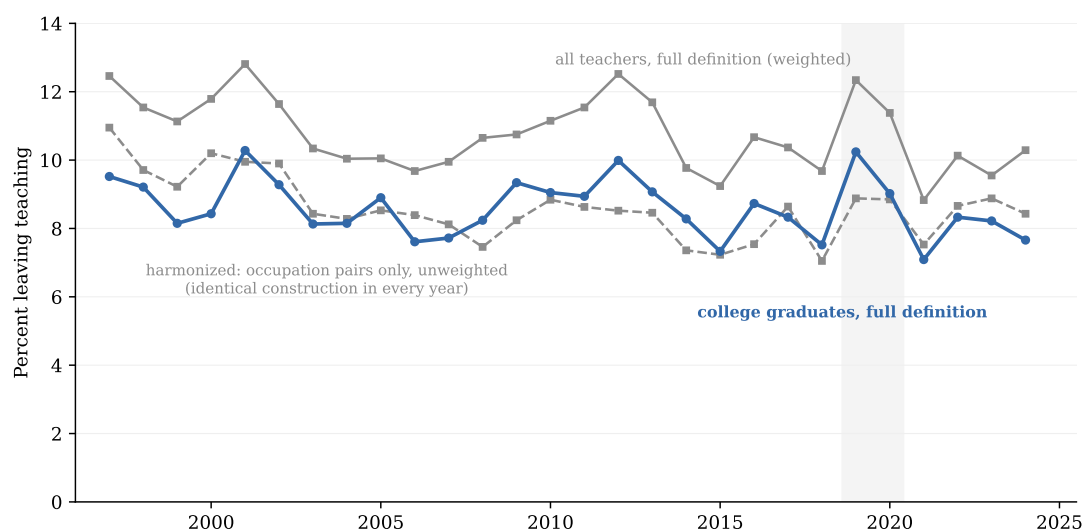


Figure 18: The leaving series under alternative definitions.

The baseline route-based college-graduate series alongside the all-education series and the harmonized occupation-pair-only construction. Levels differ by universe; profiles coincide. Source: author's calculations from March CPS supplements.

Table 5: Robustness of the attrition model.

	(1) Probit	(2) LPM, wtd.	(3) Logit	(4) LPM, FE	(5) Probit, full	(6) LPM, pair
Age (years)	-0.80*** (0.08)	-1.40*** (0.13)	-0.68*** (0.07)	-1.38*** (0.13)	-0.83*** (0.05)	-1.28*** (0.13)
Age ² /100	0.96*** (0.08)	1.71*** (0.15)	0.84*** (0.07)	1.69*** (0.15)	0.99*** (0.05)	1.53*** (0.14)
Female	-1.42*** (0.34)	-1.68*** (0.45)	-1.35*** (0.34)	-1.61*** (0.45)	-1.34*** (0.21)	-1.49*** (0.42)
Black	1.86*** (0.49)	2.22*** (0.69)	1.77*** (0.49)	2.39*** (0.72)	2.10*** (0.31)	1.44** (0.63)
Married	-0.23 (0.47)	0.49 (0.58)	-0.23 (0.49)	0.47 (0.59)	-0.32 (0.29)	0.46 (0.54)
Separated/divorced/widowed	0.96 (0.59)	1.78** (0.78)	0.97 (0.59)	1.73** (0.78)	0.29 (0.36)	1.56** (0.73)
Master's degree or higher	-0.84*** (0.30)	-0.50 (0.38)	-0.85*** (0.30)	-0.45 (0.40)	-0.58*** (0.18)	-0.59* (0.35)
Public school	-1.74*** (0.32)	-1.56*** (0.47)	-1.64*** (0.32)	-1.59*** (0.48)	-1.28*** (0.20)	-1.24*** (0.44)
Pension plan at work	-2.46*** (0.32)	-2.57*** (0.45)	-2.47*** (0.32)	-2.56*** (0.45)	-3.11*** (0.20)	-2.08*** (0.42)
Part-time (<35 h/wk)	4.37*** (0.37)	8.25*** (0.85)	3.95*** (0.35)	8.22*** (0.85)	4.62*** (0.22)	6.82*** (0.79)
Worked full year (50+ wks)	-9.26*** (0.32)	-12.47*** (0.54)	-9.66*** (0.33)	-12.55*** (0.54)	-10.32*** (0.19)	-9.77*** (0.49)
Number of children	-0.47*** (0.17)	-0.38* (0.20)	-0.65*** (0.18)	-0.37* (0.20)	-0.72*** (0.11)	-0.33* (0.18)
Child under 6	0.45 (0.51)	0.50 (0.50)	0.52 (0.54)	0.54 (0.51)	0.62** (0.30)	0.67 (0.47)
New baby this year	4.61*** (0.78)	5.42*** (1.23)	4.85*** (0.77)	5.44*** (1.23)	4.76*** (0.43)	4.78*** (1.17)
Observations	28,900	28,900	28,900	28,900	85,430	28,900

Dependent variable: leaving teaching (percentage points). Column (1): baseline probit AMEs, 2015–2024, year effects, unweighted. (2): weighted linear probability model, HC1. (3): logit AMEs. (4): weighted LPM with state and year effects. (5): probit on the full 1997–2024 window. (6): weighted LPM with the occupation-pair definition of leaving. Standard errors in parentheses; *, **, *** denote 10, 5, and 1 percent.

Table 6: The attrition model in subsamples.

	Women	Men	Public	Private	Under 45	45+
Age (years)	-1.35*** (0.15)	-1.41*** (0.28)	-1.38*** (0.16)	-1.44*** (0.25)	-0.00 (0.51)	-1.31** (0.59)
Age ² /100	1.66*** (0.17)	1.69*** (0.30)	1.77*** (0.18)	1.61*** (0.26)	-0.22 (0.73)	1.65*** (0.51)
Female	—	—	-0.71 (0.48)	-4.06*** (1.03)	-1.20** (0.59)	-2.05*** (0.70)
Black	2.45*** (0.77)	1.20 (1.48)	2.01*** (0.75)	2.41 (1.47)	3.19*** (1.00)	0.74 (0.91)
Married	0.18 (0.66)	1.72 (1.23)	0.81 (0.64)	-0.07 (1.24)	0.62 (0.70)	-0.29 (1.05)
Separated/divorced/widowed	1.43 (0.88)	2.94* (1.76)	2.25** (0.88)	0.46 (1.58)	1.34 (1.05)	1.07 (1.21)
Master's degree or higher	-0.28 (0.43)	-1.51* (0.83)	-0.53 (0.41)	-0.48 (0.81)	-1.37*** (0.48)	0.13 (0.59)
Public school	-1.12** (0.53)	-3.11*** (1.04)	—	—	-3.53*** (0.60)	0.61 (0.74)
Pension plan at work	-1.95*** (0.50)	-4.54*** (1.01)	-2.99*** (0.54)	-2.03** (0.80)	-2.50*** (0.56)	-2.74*** (0.71)
Part-time (<35 h/wk)	7.86*** (0.93)	10.63*** (2.07)	10.96*** (1.21)	5.37*** (1.21)	7.96*** (1.17)	7.68*** (1.20)
Worked full year (50+ wks)	-12.35*** (0.60)	-13.01*** (1.20)	-10.33*** (0.59)	-16.85*** (1.11)	-8.29*** (0.69)	-16.89*** (0.82)
Number of children	-0.46** (0.23)	0.03 (0.39)	-0.10 (0.22)	-0.92** (0.42)	-0.41 (0.26)	-0.52 (0.33)
Child under 6	1.30** (0.59)	-2.22** (0.97)	0.97* (0.52)	-0.35 (1.19)	-0.07 (0.59)	-1.28 (1.46)
New baby this year	5.67*** (1.49)	5.22** (2.17)	4.69*** (1.30)	6.51** (2.73)	5.54*** (1.25)	13.64 (9.34)
Observations	22,114	6,786	20,648	8,252	15,384	13,516

Weighted linear probability models with year effects, 2015–2024, HC1 standard errors in parentheses; “—” marks regressors without variation in the subsample. *, **, *** denote 10, 5, and 1 percent.

Table 7: Paid family leave and the birth-year exit.

	Coefficient (pp)	SE	<i>p</i>
New baby	+6.60	(1.19)	0.000
Paid family leave active	+0.47	(0.35)	0.181
New baby × paid leave	−2.10	(2.24)	0.349

Weighted LPM of leaving on a new-baby indicator interacted with an indicator for the state having an active paid family leave program (first full benefit years: CA 2005, NJ 2010, RI 2014, NY 2018, WA 2020, DC/MA 2021, CT 2022, OR 2024), with state and year effects and demographic controls; standard errors clustered by state. $n = 77,883$ teacher-years; 268 births under active programs. The interaction is right-signed and uninformative: the design awaits more treated cohorts.